Homework 1 ALU (Arithmetic Logic Unit) Design

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Date: 2025/09/15

EE

LPHPLAB
VLSI Design LAB

Introduction

- The Arithmetic Logic Unit (ALU) is a fundamental building block of a computer processor.
- In Homework1, you are required to design and implement a 32-bit ALU supporting selected RV32IM instructions.
- Your ALU should correctly perform arithmetic, logical, shift, relational, and multiplication operations as specified.
- The goal of this assignment is to:
  - Understand instruction decoding in RISC-V.
  - Implement multiple ALU functionalities in Verilog.
  - Verify your design through simulation and waveform analysis.

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Hardware description (1/2)_I/O Port Description

Signal name	I/O	Width	Description
funct3	In	3 bits	RISC-V instruction field used to select ALU operation
funct7	In	7 bits	RISC-V instruction field used to differentiate operations (e.g., ADD vs SUB, SRL vs SRA)
src_A_i	In	32 bits	First source operand (corresponds to rs1)
src_B_i	In	32 bits	Second source operand (corresponds to rs2)
result_o	Out	32 bits	ALU output result

- In Homework1 consider **src_A_i = $(rs1), src_B_i = $(rs2), result_o = $(rd)**
- **Note:** [rs1: register number , $(rs1): data of register (rs1)]
- For R-type & M Standard instruction format:

31 25	24 20	19 15	14 12	11 7	6 0
funct7	rs2	rs1	funct3	rd	opcode

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Hardware description (2/2)_Instruction(RV32IM Subset)

	Instruction	funct7	funct3	Operation Description
0	ADD	0000000	000	rd = rs1 + rs2
1	SUB	0100000	000	rd = rs1 - rs2
2	SLL	0000000	001	rd = rs1 << (rs2[4:0])
3	SLT	0000000	010	rd = (rs1 < rs2) ? 1 : 0 (signed)
4	SLTU	0000000	011	rd = (rs1 < rs2) ? 1 : 0 (unsigned)
5	XOR	0000000	100	rd = rs1 ^ rs2
6	SRL	0000000	101	rd = rs1 >> (rs2[4:0]) (logical)
7	SRA	0100000	101	rd = rs1 >>> (rs2[4:0]) (arithmetic)
8	OR	0000000	110	rd = rs1 | rs2
9	AND	0000000	111	rd = rs1 & rs2
10	MUL	0000001	000	rd = (rs1 * rs2)[31:0]
11	MULH	0000001	001	rd = (signed(rs1) * signed(rs2))[63:32]
12	MULHSU	0000001	010	rd = (signed(rs1) * unsigned(rs2))[63:32]
13	MULHU	0000001	011	rd = (unsigned(rs1) * unsigned(rs2))[63:32]

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Commands

Tool		Commands
VCS	Compile & simulate	% vcs -R –sverilog tb_ALU.sv -debug_access+all -full64
	Debug with waveform	% vcs -R –sverilog tb_ALU.sv -debug_access+all -full64 +define+FSDB

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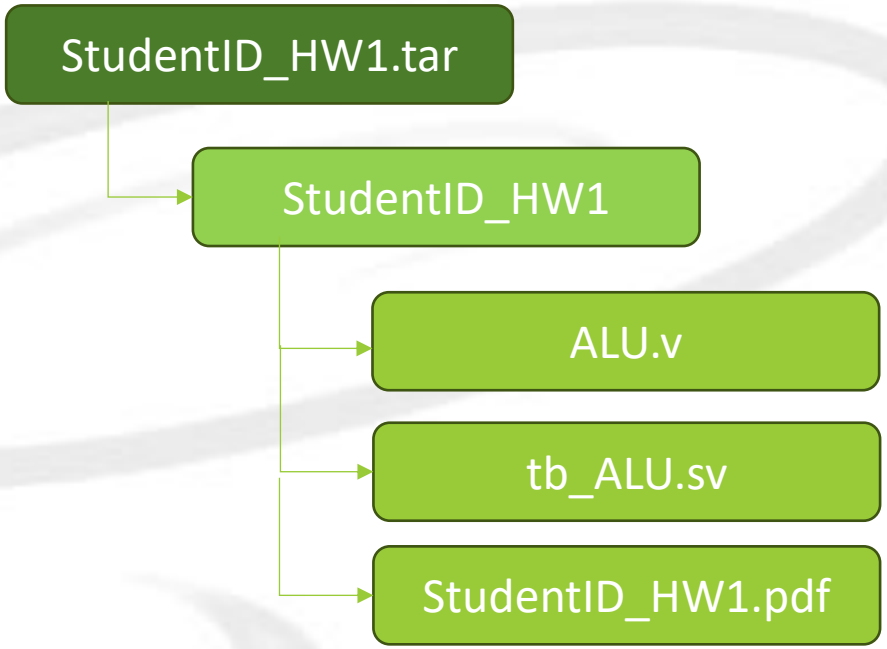
Requirement & File format (1/2)

- You must finish ALU.v and pass all patterns.
- Complete the report (in PDF format).
- Homework submission must follow the file hierarchy!
- Deadline:2025/10/06,15:00



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Requirement & File format (2/2)

Report Content:

- 1. Design principles (including data path diagram).
- 2. Simulation results (terminal screenshots & waveforms & explanation).
- 3. Superlint coverage (Superlint screenshots & coverage percentage)
- 4. Discussion of problems & solutions (optional).
- Must be typed and clear – no handwritten reports.
- Do NOT include Verilog code in the report.

Submission Rules

- File/folder names must be English only, no Chinese characters.
- Do not upload waveform dump files (e.g., .vcd, .fsdb).
- Do not upload unnecessary large files.

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Grading policy (100%)

- Simulation pass (70%) (Compile errors = 0 credit)
- Superlint coverage > 70% (10%)
- Report (20%)

- Late submission within 24hr → score × 0.8
- Late submission within 48hr → score × 0.6
- Late submission within 72hr → score × 0.4
- Late submission more than 72hr → assignment will NOT be accepted. (score × 0)
- Submission that fails to follow the file hierarchy and format will result in a 10-credit reduction.

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Friendly reminder

- Important: The TA will also use their own testbench (including hidden test cases) for grading. Your design must be correct and robust enough to handle all possible inputs.
- Please complete the assignment on your own; discussion with peers is recommended, but do not copy.
- Please make sure that your code can be compiled in VCS.
- Please submit your work according to the specified format.

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Thanks for Listening!

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